

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2017

Programme: BE

Full Marks: 100

Course: Computer Graphics

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What do you understand by computer graphics? Mention some of the advantages of computer graphics. 2+5
b) Explain the working principal of LCD and LED. 4+4
2. a) Explain the techniques of pixel considered as connected? Write logic to draw a circle using midpoint circle algorithm. 3+5
b) Using the Bresenham's line drawing algorithm predict the pixels on the line from (2,2) to (12,10). 7
3. a) Show that the composition of two successive rotation are additive. 5
b) Derive the composite transformation matrix for reflection of an object about a line $Y=mx+c$. Apply the derived matrix for the object A (4,2) B(7,3) C(9,2) D(10,1) on to the line $y=3x$. 10
4. a) What are the issue in 3D that makes it more complex than 2D? Derive an equation for 3D translation and reflection. 3+4
b) Define Projection. Difference between parallel and perspective projection along with an equation. 2+6
5. a) Compare object space method with image space method. Explain scan line algorithm for detecting visible surfaces with suitable figure. 4+4
b) What is Specular reflection? Explain the total intensity due to Specular reflection. 2+5
6. a) Explain Gouraud Shading and Phong Shading technique in detail with their advantage and disadvantage. 4+4
b) Define Graphics file format. Explain with example, the need for machine Independent Graphical Language. 2+5
7. Write short notes on: (Any two) 2x5
 - a) Frame Buffer Organization
 - b) Beizer Curve
 - c) Depth Buffer method
 - d) Cohen-Sutherland

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Graphics

Semester: Spring

Year : 2017
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Compare and contrast raster scan display and vector scan display architecture. 7
- b) Define Display controller? What are the major application areas of computer graphics? 8
2. a) Define resolution. Suppose RGB raster system to be designed using on 8 inch x 10 inch screen with a resolution of 100 pixels per inch in each direction. If we want to store 6 bits per pixel in the frame buffer, how much storage (in bytes) do we need for frame buffer? 8
- b) Digitize one octant of a circle by using midpoint circle generation algorithm center at origin and radius is 12. 7
3. a) Derive an equation for drawing a line using Bresenham's algorithm for slope less than one. 8
- b) Explain two dimensional line clipping algorithm with suitable example. 7
4. a) Differentiate between windows and viewport? Explain the steps of viewing transformation. 7

OR

- A mirror is placed vertically such that it passes through the points (10,0) and (0,10). Find the reflected view of triangle ABC with coordinates A(5,50), B(20,40) and C(10,70).
- b) Describe the rotation of an object about an axis, which is parallel to any of three coordinate axes of coordinate system. 8
 5. a) Explain the back face detection method with an example. 8
 - b) What is ambient light? Compare diffuse reflection with Specular reflection. 7

6. a) Explain Fast Phong shading algorithm in detail with necessary equations and figures. 7
b) Why machine independent programming language is used? Discuss about OPENGL. 8
7. Write short notes on: (Any two) 2x5
a) 2D rotation
b) Graphics file format
c) RGB color model

7

8

8

7

8

7

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POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Graphics

Semester: Fall

Year : 2018
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate between raster and random scan system with their corresponding architectures. 7
b) Define resolution and persistence. How long would it take to load a 640×480 frame buffer with 12 bits per pixel, if 10^5 bits can be transferred per second? How long would it take to load 24 bits per pixel frame buffer with a resolution of 1280×1024 with using the same transfer rate? 8
2. a) How decision parameter is calculated in Mid point circle method. Show all necessary derivation. 7
b) Explain boundary fill and flood fill algorithm in detail. 8
3. a) What will be the final coordinates of a triangle with vertices A(2,3) B(3,3) C(3,2) after reflecting it about the line $y = x$? 8
b) What do you mean by windowing and Clipping? Explain window to viewport transformation pipeline. 7
4. a) Show how to use a 3 Dimensional matrix to rotate a unit cube about the axis defined by vector (1,1,1). 8
b) Explain Bezier method of curve drawing. 7
5. a) Define the terms: depth cueing and surface rendering. Write down the necessary algorithms for any one of the image space method. 8
b) How phong shading is different from Gouraud shading? Explain it. 7
6. a) Obtain a illumination equation due to ambient, diffused and specular reflection model at a point. 8
b) Explain the need for machine independent graphical languages. And also explain about GKS. 7
7. Write short notes on: (Any two) 2×5
 - a) Application areas of computer graphics
 - b) Working principle of LED
 - c) Composite transformations

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Graphics

Semester: Spring

Year : 2018
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain frame buffer? How is computer graphics applicable in the field of GUI, Entertainment and medical science? Explain 5
b) Calculate the access time for a pixel and a row for a graphics system having resolution of 1024×640 and frequency of 60 Hz. 5
c) Explain raster scan system with video controller. 5
2. a) How colors are displayed in monitor? 5
b) Explain in steps the Z-buffer algorithm. 5
c) Explain scan Line Method. 5
3. a) Derive an equation for calculating points of an ellipse. 7
b) Rasterize the points of given line end points A(-2 , -4) and B(-6,-9) using Bresenham's line drawing algorithm. 8
4. a) What is windowing and clipping? Derive window to viewport transformation matrix. 7
b) Apply Cohen Sutherland line clipping algorithm for calculating the saved portion of a line from (2,7) to (8,12) in a window ($X_{wmin} = Y_{wmin} = 5$ and $X_{wmax} = Y_{wmax} = 10$) 8
5. a) Define Projection? Derive a matrix for a parallel projection. 7
b) Calculate (x, y) coordinate of Bezier curve described by the following 4 control points (0, 0), (1, 2), (3, 3), (4, 0). Assume any needed values. 8
6. a) Explain the Gouroud shading method with its advantages. 5
b) Explain why is RGB called as additive and CMYK called as subtractive model? 5

c) Explain open GL.

5

7. Write short notes on: (Any two)

2×5

a) Explain shading method of intensity interpolation.

b) Explain different file formats.

c) Viewing in 3D

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Graphics

Semester: Fall

Year : 2019
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Explain the use of computer graphics emphasizing the application of graphics in the field of entertainment. 8
- b) Consider a non-interlaced raster monitor with a resolution of 1280x1024. If horizontal and vertical retrace times are 20 microsecond each, than calculate the fraction of the total refresh time per frame spent in retrace of the electron beam? Assume refresh rate of 60 frames per second. 7
- a) Define Video Controller? Differentiate between Beam penetration and shadow mask method? 8
- b) Explain the working of DDA line drawing algorithm with suitable examples. Write its advantage and disadvantage. 7

OR

Explain Symmetrical property of circle. Write midpoint circle algorithm and apply that algorithm to find the pixel values of the circle whose radius $r = 10$ and centre of the circle $= (0, 0)$.

- a) Define Decision Parameter in Bresenham's line drawing? Digitize a circle $(x-2)^2 + (y-3)^2 = 25$ using a midpoint circle drawing algorithm. 8
- b) Determine window to viewport transformation matrix for window (5, 10) (15, 20) and for viewport (8, 12) (12, 18). Note the coordinates values are for lower left and upper right corner. 7
- a) Why do you need clipping? Explain the Cohen Sutherland line clipping algorithm. 8
- b) Derive the composite matrix for reflection an object about an arbitrary axis in 3D Space. 7
- a) Explain and derive transformation matrix of 3D rotation about a line 8

not parallel to any one axis.

- b) Distinguish between Image space and Object space method. How A-buffer method removes the drawback of Z-buffer method. 7
- 6. a) What do you mean by ambient light? Compare between Additive and subtractive color model. 7
- b) Define OPenGL? Explain the different file format used in Graphics to save image. 8
- 7. Write short notes on: (Any two) 2×5
 - a) Pros and Cons of Vector Graphics
 - b) A-Buffer Method
 - c) Need for Machine Independent Graphical Languages.

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Graphics

Semester: Spring

Year : 2019
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is computer Graphics? How it is used in education and training and entertainment? 8
b) Explain the architecture of raster scan system with importance of video controller. 7
2. a) Explain the Bresenham's Line drawing algorithm with suitable example. 8
b) Derive mid-point Circle algorithm. 7
3. a) What is clipping? Explain Cohen Sutherland's line clipping algorithm with suitable example. 8
b) Explain 2D transformation using Homogeneous coordinator system. 7
4. a) Explain Beizer curve and also specify the properties of Beizer curve 8
b) Explain and derive parallel projection transformation matrix. 7
5. a) What is Hidden surface? Explain the back face detection method 7
b) Explain Gouraud and phong shading method with its advantages and disadvantages. 8
6. a) Explain ambient, diffuse and specular reflection. 7
b) Explain about PHIGS and GKS language. 8
7. Write short notes on: (Any two) 8
a) Touch panel
b) Area Subdivision Method
c) OPENGL 2x5

POKHARA UNIVERSITY

Level: Bachelor
Programme: B E
Course: Computer Graphics

Semester: Fall

Year : 2020
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) How the entertainment and gaming industry has revolutionized by the advancement in computer graphics explain your answer with some real life examples. 7
b) Discuss the difference between raster and random scan display system with its architectural diagram. 8
2. a) Derive an equation for calculating points of an ellipse. 7
b) Rasterize the points of given line end points A (-2 , -4) and B (-6,-9) using Brenham's line drawing algorithm. 8
3. a) Find the Transformation matrix for window to viewport Transformation. 7
b) What is windowing and clipping; how a polygon can be clipped explain? 8
4. a) Derive equation for Bezier curve in quadratic polynomial and specify the blending function 7
b) Define projection. Difference between parallel and perspective projection with figure 8
5. a) Explain the importance of hidden surface removal in computer graphics, explain scan line method of hidden surface removal. 7
b) Differentiate between Gouraud and phong shading with algorithm. 8
6. a) What is color model. Explain RGB and CMYK color model. 7
b) Explain GKS and PHIGS. Also list out the available graphical file format. 8

7. Write short notes on: **(Any two)**

2×5

a) Light pen

b) A-Buffer Method

c) Need of illumination Model

POKHARA UNIVERSITY

Level: Bachelor

Semester -Spring

Year: 2020

Program: BE

Full Marks: 70

Course: Computer Graphics

Pass Marks: 31.5

Time: 2 hrs.

Candidates are required to answer in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

Group - A: (5×10=50)

- Q. N. 1 Why graphical system contains Digital-to-Analog Converter (DAC) and what is the role of frame buffer organization? 8 way symmetry is not applicable for Ellipse drawing mechanism. Why? Derive the initial decision parameter for Mid-point circle drawing algorithm. 10

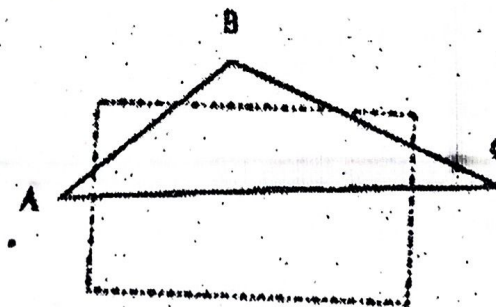
OR

Throughout the first quadrant dividing it into two parts, the regions can be formed by considering the slope of the curve. Assume the slope of curve is less than one then we are in region one and when the slope becomes greater than -1 then in region 2. Considering major and minor axis throughout the first quadrant. What would be appropriate method used to draw the elongated part of circle

- Q. N. 2 What is the importance of window and viewport transformation, explain with its practical application? Why we used to homogenous coordinate system in transformation, is it true that without homogenous coordinate system we could not have the final transformation matrix while performing animation? 10

- Q. N. 3 Rotate a triangle A(5,5) B(10,10) and C(@,@) by 45 in CW and scale to half of its original size about origin where @ is the last digit of your exam roll number. 10

- Q. N. 4 Why clipping is necessary in computer graphics? Clip the given polygon using Sutherland-Hodgeman Algorithm and explain every step considered. 10



- Q. N. 5 If you ask people to describe a color in terms of 24 bit color model, color are represented by varying intensities of different light and if the maximum intensities of color is used. Which color model will be applicable to this perceptual organization of color? C is sometime called minus R, M is sometimes called minus G and Y is sometime called minus B. How would you represent this model geometrically 10

Group - B: (1×20=20)

- Q. N. 6 To develop a gaming application which process model that you are going to recommend as a software project manager when the requirement are very hard to collect and not stable. During the development of the application which data structure can be suitable for you application and in the final rendering process which rendering model that you are going to suggest if we have enough computing resources where we are aiming to develop the seamless finest graphics ?

SHIVAM STATIONERY & COMPUTER REPAIRING CENTRE

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ADDRESS : THAPA CHOWK

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Graphics

Semester: Fall

Year : 2021
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define computer graphics. List the application of computer graphics in different fields. 7
b) In a true color system having resolution of 1024×768 having the refresh rate of 60fps calculate the following 8
 - i. Size of frame buffer
 - ii. Access time of one frame
 - iii. Access time for one pixel
 - iv. Access time for one row
- Note: convert your memory into Mega Byte.
2. a) Differentiate between raster and Vector scan display system along with their architecture. 8
b) Digitize the first octant of a circle having radius $r=8$ and centered at (3,4) 7
3. a) Prove that successive translation and rotation is additive. 8
b) Explain the role of composite transformation in 2D/3D geometric transformation. Explain viewing pipelining in 2D. 7
4. a) What is 3D transformation? Rotate the triangle A(0,0), B(2,2), C(4,2) about the origin by an angle of 45° . 8
b) How you represent different objects in 3D. Differentiate between parallel and perspective projection with example? 7
5. a) What is Mach band effect? Differentiate between Gouraud and Phong shading. 8
b) Define color model in computer graphics. Differentiate between additive color and subtractive color. 7

6. a) Explain the importance of hidden surface removal in computer graphics. What are the drawbacks of z-buffer method and how it is corrected in A-buffer? 7
- b) Explain how machine independent graphical language are more preferable to develop graphical project. 8
7. Write short notes on: (**Any two**) 2×5
- a) Open GL
 - b) Beizer curve
 - c) Polygon Table

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Graphics

Semester: Spring

Year : 2021
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Compare and contrast the Raster and Random system with the necessary diagram. 7
- b) Define resolution and persistence. Explain the digital to analog converter in frame buffer organization? 8
2. a) Write Bresenham's line drawing algorithm along with necessary derivation for positive slope less than 1 i.e. $|m| < 1$. Trace the algorithm for the line with end points A(6,12) and B(10, 15). 8
- b) Explain the Sutherland Hodgeman polygon clipping algorithm considering the four different cases. 7

OR

Let R be the rectangular window whose lower left hand corner is at L (-3, 1) and upper right-hand corner is at R (2, 6). Use Cohen – Sutherland algorithm to clip the line segments A (-4, 2) and B (-1, 7).

3. a) Derive a composite transformation matrix for reflecting an object about a line $y=x+4$ in 2D. 7
- b) Differentiate between window and viewport. Derive a matrix for window to viewport transformation. 8
4. a) What do you mean by a projection? Derive expression to obtain the perspective projection of any arbitrary point. 7
- b) Why filling algorithm is required in computer graphics? Explain about scan line polygon filling algorithm. 8
5. a) What are the different ways of representing 3D objects in computer Graphics? Explain how can you represent a polygon surface as a 3D-object. 8
- b) Why is it necessary to detect visible surfaces, in case of 3D viewing? 7

Explain Z-buffer Algorithm for hidden surface removal.

6. a) Describe how Gouraud Shading algorithm can be used in rendering a realistic 3D object. 7
- b) Why do we need the machine independent graphical languages? List out some of the graphical file formats and explain them in short. 8
7. Write short notes on: (**Any two**) 2x
 - a) Diffuse and specular Reflection
 - b) Color models
 - c) Application of computer graphics

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Graphics

Semester: Fall

Year : 2022
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Computer Graphics? Explain the applications of CG in Computer Simulation, Scientific Visualization and CAD. 7
b) Define Refresh rate and Resolution. Consider a raster scan system having 20-inch by 30-inch screen with a resolution of 300 pixels per inch in each direction. If the display controller of the system refreshes the screen at a rate of 60 frames per second, how many pixels could be accessed per second and what is the access time per pixels of the system. 8
2. a) Define CRT? Explain with neat diagram about the working principle of shadow mask method? 7
b) Consider a line from (2,1) to (8,3) using DDA algorithm to rasterize a line. 8
3. a) Explain the boundary fill algorithm in detail. How this approach differs from flood fill? 8
b) Prove that: 7
 - i. Two successive Translations are Additive
 - ii. Two successive Scaling are Multiplicative
4. a) Window port is given by (100,100,300,300) and viewport is given by (50,50,150,150). Convert the window port coordinate (200,200) to the viewport coordinate. 7
b) A mirror is placed vertically such that it passes through the points (5, 0) and (0, 5). Find the reflected view of triangle ABC with coordinates A (5, 30), B (30, 50) and C (20, 60). 8
5. a) How do you represent 3D objects by using Polygon Tables? How is the consistency of geometric data table checked and what are the rules for generating error free polygon tables? 7

- b) What is OpenGL? Why GLUT is implemented in OpenGL Explain Callback function. 8
6. a) Derive the equation for cubic Bezier curve and find the coordinate at $t=0.2$ with respect to the control points (1, 1), (4, 6) (8,-3) and (12, 2). 8
- b) Why depth sorting method is called Painter's Algorithm? Explain scan line method for visible surface detection with an example. 7
7. Write short notes on: (**Any two**) 2×5
- a) Light pen
 - b) Color models
 - c) Phone shading

Level: Bachelor
Programme: BE
Course: Computer Graphics

POKHARA UNIVERSITY
Semester: Spring

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define computer graphics with its type. How has the evolution of computer graphics technology impacted the gaming industry in terms of realism and player immersion? 8
b) Graphical systems are composed of DAC and Frame Buffer organization. Why? Explain the working principle of Data gloves with example. 7
2. a) How much time is spent scanning across each row of pixels during screen refresh on a raster system with a resolution of 1280 by 1024 and a refresh rate of 60 frames per second? 7
b) Why Circle and Ellipse uses different point symmetry? Explain the mid-point circle generation algorithm by calculating its initial decision parameter. 8
3. a) What is uniform scaling? Reflect an object (2, 3), (4, 3), (4, 5) about line $y = x + 1$. 8
b) How is Scan line algorithm different from Flood fill algorithm for filling polygons? Explain. 7
4. a) What do you understand by the term clipping? Explain the Cohen-Sutherland line clipping algorithm with suitable example. 8
b) Why we need the machine independent graphical language? Explain two different graphical software standards. 7
5. a) Define blobby objects with example. What is cubic Bezier curve? Derive the equation of Cubic Bezier curve. 8
b) Why is it required to detect visible surfaces in 3D viewing and not in 2D? How is Back face detection approach different from Z Buffer approach for detecting visible surfaces? 7

6. a) Explain the 3D viewing pipeline. Briefly explain the significance of World coordinate system, Viewing coordinate system, Normalized viewing coordinate system and Device coordinate system in the viewing pipeline. 8
- b) What is illumination model? Differentiate between Phong Shading and Gouraud Shading. 7
7. Write short notes on: (Any two) 2x5
- a) Specular Reflection
- b) Parallel vs. Perspective Projection
- c) Area Subdivision Method

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Graphics (Old)

Semester: Fall

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate between raster scan and vector scan system with its architectural diagram. 7
- b) How is Resolution different from Aspect ratio? A raster scan system has 14 inch by 10 inch screen with a resolution of 220 pixels per inch in each direction. What would be the access time per pixel of the screen if the video controller refreshes the screen at a rate of 65 frames per second? 8
2. a) Define Emissive and Non-Emissive Display. Explain the working principle of LCD display. 7

OR

Consider a line from (5, 5) to (13, 9). Use Bresenham's algorithm to rasterize the line.

- b) Explain flood fill algorithm using four connected method. Differentiate between boundary and flood fill algorithm. 8
3. a) Find the final coordinates of a triangle with vertices A(5,8) B(15,8) C(10,16) after reflecting it about line $y = -2x - 2$. 8
- b) Window is defined as (20,20),(90,20),(90,70),(20,80). Use Cohen Sutherland line clipping algorithm to clip the line P(10,30) Q(80,90). 7
4. a) Differentiate between Parallel and perspective projection with an example. 7
- b) How is ISM approach different from OSM approach for detecting visible surfaces? How does Backface detection algorithm work for detecting visible surfaces? 8
5. a) The 3D viewing process is more complex than 2D viewing process. Why? How the mismatch between 3D and 2D is compensated. Explain the steps with its appropriate figure. 7
- b) What is cubic Bezier curve? Derive the equation of cubic Bezier curve. 8
6. a) Why do programs that use OpenGL can be run on multiple hardware platforms? Explain the basic OpenGL API that are used for rendering graphical objects. 7
- b) Explain the color models with its geometrical representation and their relationship. 8

2×5

7. Write short notes on: (Any two)

- a) PHIGS
- b) Data Gloves
- c) Diffuse Reflection

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Graphics

Semester: Spring

Year : 2024
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What are the basic components of graphical System? list out the areas of application of computer graphics by emphasizing your answer on CAD. 7
- b) What role does a frame buffer play in graphical rendering? A raster graphics system can produce a total number of 16777216 different level of intensities out of a single pixel on the screen, if the total resolution of the screen is 1280x800, what is the required size of the frame buffer of this system? 8
2. a) Write the algorithm for both condition of Bresenham's line drawing algorithm and derive the initial decision parameter for less than 1. 8
- b) Why Circle and Ellipse uses different point symmetry? Digitize the circle with radius 10. 7
3. a) What will be the final coordinates of a triangle with vertices A(5,8), B(15,8), C(10,16) after rotating it about any fixed point (10,5) and then shifting it down by two units? 8
- b) Define window and viewport. Derive a matrix for windows to viewport transformation along with necessary steps. 7
4. a) Why is it required to map object description from the world coordinates to the device coordinates? Explain the mapping process with the necessary equations. 8
- b) Differentiate between Parallel and perspective projection with real life application example. 7
5. a) Why depth sorting method is called painter's algorithm? Explain scan line method for visible surface detection with an example. 7
- b) Define Specular reflection. Calculate the total intensity due to specular reflection caused by ambient light using phong model. 8
6. a) Compare Constant, Gouraud and Phong shading. Also conclude the best model for shading and why? 8

- b) Why we need the machine independent graphical language? Explain the rendering pipeline of OpenGL.
7. Write short notes on: (**Any two**)
- a) Raster vs. Random Architecture
 - b) 3D Representations
 - c) Boundary vs. Flood Fill Algorithm

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Graphics

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define computer graphics. Explain the use of computer graphics in Engineering education, Simulation, and Cartography with an example. 7
b) What is flat panel display? Explain the working principles of shadow mask method with figure. 8
2. a) A colour CRT monitor capable of displaying 1000 different colours having refresh rate of 50hz and resolution of 640 x 480 calculate the following: 8
 - i. Size of frame buffer
 - ii. Time required to access one frame
 - iii. Time required to access one row
 - iv. Time required to access one Pixel.
- b) Why Bresenham's algorithm is considered more efficient than DDA algorithm. Write the algorithm for both condition of Bresenham's line drawing algorithm. 7
3. a) Digitize one octant of a circle by using midpoint circle generation algorithm center at (5, 10) and radius is 12. 7
b) Derive the transformation matrix for reflection about the line $y=mx+b$. 8
4. a) Use Cohen-Sutherland algorithm to clip the line P1 (40, 15) and P2 (75, 45) against the window (50, 10) and (80,40). 8
b) What is the importance of curve in Computer Graphics? Explain Cubic Bezier curve with proper derivation. 7
5. a) Compare Parallel Projection with Perspective Projection. Derive the equations of perspective projection with necessary figures. 7
b) Differentiate between ISM and OSM. Explain Scan Line method of back face detection. 8

6. a) List out the different surface shading methods and explain Gouraud shading. 8
- b) What is the significance of Portability of Software? Compare GKS and PHIGS with an example. 7
7. Write short notes on: (**Any two**) 2×5
- a) Boundary and Flood Fill Algorithm
- b) RGB color model
- c) Interlaced Refresh Procedure

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Graphics

Semester: Spring

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

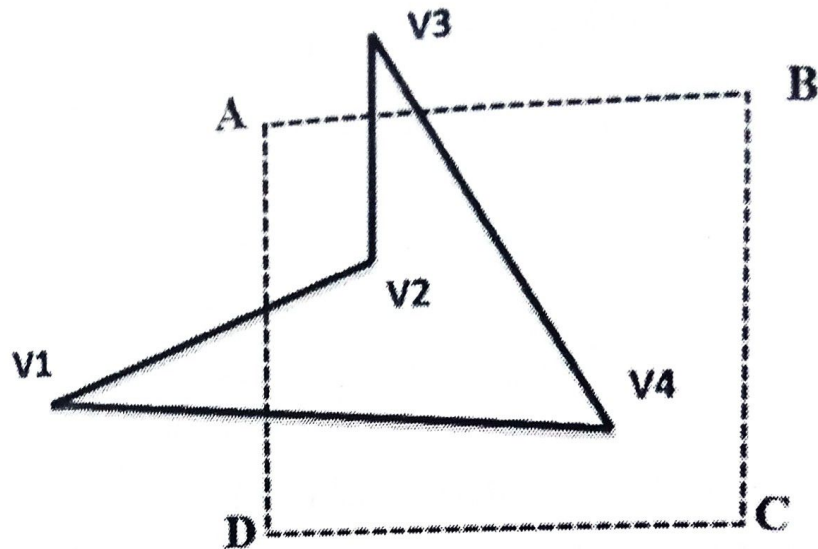
The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define computer graphics. Explain the use of Computer Graphics in creating realistic Virtual Reality based applications. 7
- b) What do you mean by hardware acceleration? Differentiate between Raster and Random scan display with its architectural diagram 8
2. a) In Case two raster systems with resolutions of 640 by 480 and 1024 by 600, how many pixels could be accessed per second in each of these systems by a display controller that refreshes the screen at a rate of 75 frames per second? What is the access time per pixel in each system? 8
- b) What role does bits in a frame buffer play in glowing the pixels on the screen with different intensities? Explain with an example. 7
3. a) Derive the algorithm for digitizing a circle with a radius of 'r' pixels with the starting pixel located at (0,r) and moving in an anticlockwise direction. 8
- b) What is scan conversion? Write the Bresenham's line drawing algorithm for slope less than 1 and derive its initial decision parameter. 7
4. a) What will be the final coordinates of a polygon with vertices A(3,4) B(5,4) C(5,2) D(3,4) after it is reflected about a line $y=2x+1$? 8
- b) Use Cohen-Sutherland algorithm to clip the line P1 (70, 20) and P2 (100, 10) against the window (50, 10) and (80,40). 7

OR

Clip the given polygon by using Sutherland-Hodgeman polygon clipping algorithm.



5. a) How can the visibility of objects be determined using object space method? Explain its advantages over image space methods.
- b) Explain the process of clipping 3D objects and mention how it differs from 2D clipping.
6. a) Explain Depth Buffer algorithm for visible surface detection. Also list its pros and cons.
- b) Explain constant shading method and mention its limitation. Propose and explain the method to overcome the limitation of constant shading.
7. Write short notes on: (**Any two**)
 - a) Graphical language
 - b) Viewing pipeline
 - c) Beizer curve

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Graphics (new)

Semester: Fall

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is Computer Graphics? Differentiate between raster and random display with its architecture. 8
- b) Why do we need hardware acceleration for graphics processing? Explain the significance of GPU in modern computer organization. Compare CPU with GPU. 7
2. a) What is scan conversion? How can equations be derived for drawing lines with slope greater than one using Bresenham's line drawing algorithm? Explain. 8
- b) What is composite transformation. Explain why do we need homogeneous coordinate system for transformation of computation in computer graphics. 7
3. a) How symmetric property of circle can be implemented to digitize circle having equation $(x-4)^2 + (y+5)^2 = 36$. Illustrate with necessary calculations using mid-point circle drawing algorithm. 8

OR

What is window and viewport? Magnify the triangle with vertices A(0,0) B(1,1) and C(5,2) to thrice its size while keeping C(5,2) fixed.

- b) Explain the Sutherland Hodgeman polygon clipping algorithm considering the four different cases with example. 7

OR

Let R be the rectangular window whose lower left hand corner is at L (-3, 1) and upper right-hand corner is at R (2, 6). Use Cohen – Sutherland algorithm to clip the line segments A (-4, 2) and B (-1, 7).

4. a) How are Two Dimensional Rotations different from Three 7

Dimensional Rotations? Explain with the matrix representations.

- b) Derive the blending function for parametric cubic curve using Hermite interpolation technique. 8
- 5. a) What are the drawbacks of Flat shading and Gouraud shading? How does normal vector interpolation scheme remove these drawbacks? 7
- b) Explain painter's algorithm for visible surface detection with its necessary test cases. 8
- 6. a) Define GKS and PHIGS. Mention different types of Graphics file format used in computer graphics. 8
- b) How OpenGL APIs can be used to maintain graphical standards? Explain. 7
- 7. Write short notes on: (Any two) 2×5
 - a) Color Models
 - b) 2D viewing pipeline
 - c) Frame Buffer

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Graphics (New)

Semester: Spring

Year : 2024
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Discuss video controller refresh operation with a neat diagram. If on average, it takes 20 ns for a video controller to access a pixel value from the frame buffer and glow a pixel on the screen, will it produce a flickering effect? (Screen resolution is 640 x 480) 7

OR

Define Computer Graphics? Explain the applications of CG in Computer Simulation, Scientific Visualization and CAD.

b) Define CRT? Explain with neat diagram about the working principle of shadow mask method? 8
2. a) What is the difference between DDA and BLA? Discuss "Midpoint Circle Drawing Algorithm" for 8 Octants. Derive an expression for decision parameter for conditions mentioned below. 7
 - i. Initial decision parameter P_0 .
 - ii. Decision parameter, when $P_k < 0$.
 - iii. Decision parameter, when $P_k > 0$.

b) Explain the boundary fill algorithm in detail. How this approach differs from flood fill? 8
3. a) Differentiate between Window and Viewport. Discuss the 2-Dimensional Viewing pipe line with different steps involved, and further derivation needed for window to viewport coordinate Transformation. 7

b) A mirror is placed vertically such that it passes through the points (5, 0) and (0, 5). Find the reflected view of triangle ABC with coordinates A (5, 30), B (30, 50) and C (20, 60). 8

OR

What is the significance of Clipping in Computer Graphics? Let ABCD be the rectangular window A(20,20), B(90,20), C(90,70) & D(20,70). Find the region code for the end points and use Cohen-

Sutherland Line Clipping Algorithm to Clip the Line P1P2 with P1(10,30) and P2(80,90).

4. a) Discuss difference between Spline approximation curve versus a interpolation curve. Derive an expression for the Perspective Projection with necessary diagram.
b) Derive the equation for cubic Bezier curve and find the coordinate at $t=0.2$ with respect to the control points (1, 1), (4, 6) (8,-3) and (12, 2).
5. a) Why depth sorting method is called Painter's Algorithm? Explain scan line method for visible surface detection with an example.
b) What is the difference between specular reflections versus diffuse reflection? Briefly explain polygon surface shading methods.
6. a) Discuss the need for machine-independent graphical languages. Explain two different graphical software standards.
b) What is the significance of Open GL in computer Graphics? Discuss open Graphics Library in details with different features associated with it.
7. Write short notes on: **(Any two)**
 - a) Raster vs Random scan system
 - b) Graphics File Format
 - c) Color Models

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Graphics (New)

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What do you mean by Computer Graphics? Explain the working principle of refresh CRT with figure. 7
- b) What are the benefits of using a GPU for graphical rendering? What makes a GPU more ideal for graphical rendering than using a CPU? 8
2. a) Mention the limitations of DDA algorithm. Derive the necessary expression of decision parameters to rasterize a line with $m < 1$. 8
- b) Digitize one octant of a circle by using midpoint circle generation algorithm center at origin and radius is 12. 7
3. a) A graphics designer needs to animate the swing of pendulum in center of display window with string attached at pivot point (350,300). If pendulum is displaced by 45 degrees from its resting position. Calculate the extreme points of bob's trajectory if string length is of 10 units. 8

OR

Reflect the triangle A(4,3), B(5,5) and C(4,6) about an axis defined by equation of line $y = 2 - 2x$.

- b) Why is it required to map an object description from a window to a viewport? A triangle with vertices A(30,30) B(50,30) C(40,35) is required to be mapped from a window with its lower left corner at (20,20) and upper right corner at (60,40) to a viewport with its lower left corner at (50,50) and upper right corner at (80,70). 7
4. a) Derive composite matrix for rotation about an arbitrary axis in 3D. 8

OR

What are non-planar surface? How can a Bezier surface be generated from Bezier curve?

- b) Rotate the triangle with vertices A(3,3), B(5,3), C(4,5) by 45° in anti clockwise direction While keeping B (3,3) fixed. 7

5. a) What role does Visible Surface detection play in displaying realistic scenes? Explain Depth Sorting approach for removing hidden surfaces. 8
- b) What is Gouraud shading? Explain it with an example. What are its drawbacks? How such drawbacks are removed? 7
6. a) What is OpenGL? Write any five GLUT call back functions with their necessary syntax. 7
- b) Why is it required to visualize data sets? Explain how data sets can be visualized? 8
7. Write short notes on: (**Any two**) 2×5
- a) 2D Viewing pipeline
- b) Perspective Projection
- c) Window to viewport transformation

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Programme: BE

Course: Computer Graphics (New)

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is a GPU and in what ways its going to assist the CPU to enhance the computer overall performance? Explain video controller with frame buffer organization. 7
- b) A monitor has 1280×1024 resolution and refreshes at 75 fps. How many pixels could be accessed per second and what is the access time per pixel of the system. 8
2. a) What are the limitations of DDA? Write the algorithm (any case) which overcomes these limitations. 7
- b) Derive the ellipse drawing algorithm for both region with initial decision parameter. 8
3. a) Why we need homogeneous coordinate system in transformation? Prove that, two successive translations are additive and two successive scaling are multiplicative in nature by using homogeneous coordinate system. 8

OR

Define window port and view port. Derive the window to viewport transformation matrix.

- b) You are provided with the clipping rectangle with coordinates A(10,10) B(10,20) C(20,10) and D(20,20). Clip a line with endpoints P(5,15) and Q(25,15) with the clipping rectangle using Cohen-Sutherland Line clipping algorithm. 7
4. a) How the rotation is performed in 3D? Write the rotational matrix in homogeneous form for X-axis, Y-axis and Z-axis Rotation. 7
- b) What is the importance of curve in Computer Graphics? Explain Cubic Bezier curve with necessary derivations. 8

OR

Differentiate between parallel and perspective projection with real life example for each projection.

5. a) Define visible surface detection algorithm. Explain accumulation buffer method for visible surface detection.
b) What is illumination model? Explain specular reflection and Phong model.
6. a) What is the significance of machine independent graphical language? Explain GKS and PHIGS.
b) Draw the architectural diagram of OpenGL and explain different callback function.
7. Write short notes on: (**Any two**)
 - a) 2D viewing pipeline
 - b) Fractal Geometry method
 - c) Color models

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